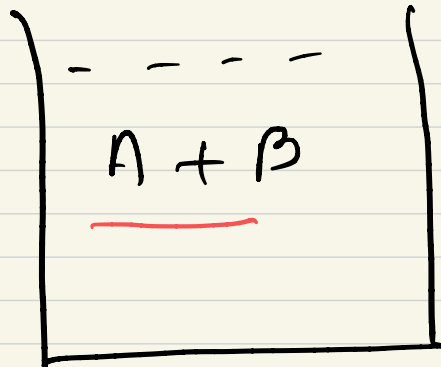


EMJ . 08/06/24

① Mixture analysis



given mass of A = w_A

" " " B = w_B ✓

molar mass of A = M_A

molar mass of B = M_B

① Percentage by weight of A

(mass = weight)

$$= \frac{\text{weight of A}}{\text{weight of A + weight of B}} \times 100$$

= weight fraction $\times 100$

(ii) % by weight of B

$$= \frac{w_B}{w_A + w_B} \times 100$$

$$= \text{weight fraction} \times 100$$

(iii) mole of A = $\frac{w_A}{M_A} = n_A$

(iv) mole of B = $\frac{w_B}{M_B} = n_B$

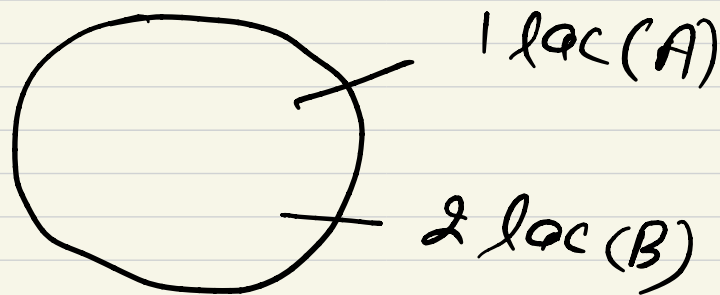
(v) mole fraction of A = $\frac{n_A}{n_A + n_B} = X_A$

$$X_A = \frac{n_A}{n_A + n_B}$$

(VII) mole fraction of B = $x_B = \frac{n_B}{n_A + n_B}$ ✓

(VIII) $x_A + x_B = 1$

$$\frac{n_A}{n_A + n_B} + \frac{n_B}{n_A + n_B} \Rightarrow \frac{n_A + n_B}{n_A + n_B} = 1$$



$$x_A = \frac{1}{1+2} = \frac{1}{3}$$

$$x_B = \frac{2}{1+2} = \frac{2}{3}$$

$x_A + x_B = 1$

(IX) mole % = mole fraction $\times 100$

$$\textcircled{\times} \% \frac{w}{w} \text{ of A} = \frac{\text{weight of A}}{\text{weight of the sol}^n} \times 100$$

Q = Solⁿ containing 4% by mass of H_2 and 96% by mass of oxygen. then find.

(i) mass of H_2

(v) χ_{H_2}

(ii) mass of O_2

(vi) χ_{O_2}

(iii) n_{H_2}

(vii) Molar mass of mixture

(iv) n_{O_2}

Solⁿ: (i) let mass of the solⁿ = 100 g

$$\textcircled{i} m_{\text{H}_2} = \frac{4}{100} \times 100 = 4 \text{ g}$$

$$\textcircled{ii} m_{\text{O}_2} = \frac{96}{100} \times 100 = 96 \text{ g}$$

$$\textcircled{\text{iii}} \quad n_{\text{H}_2} = \frac{4}{2} = 2 \text{ mol}$$

$$\textcircled{\text{iv}} \quad n_{\text{O}_2} = \frac{96}{32} = 3 \text{ mol}$$

$$\textcircled{\text{v}} \quad x_{\text{H}_2} = \frac{n_{\text{H}_2}}{n_{\text{H}_2} + n_{\text{O}_2}} = \frac{2}{2+3} = 0.4$$

$$\textcircled{\text{vi}} \quad x_{\text{O}_2} = \frac{n_{\text{O}_2}}{n_{\text{O}_2} + n_{\text{H}_2}} = \frac{3}{2+3} = 0.6$$

$$x_{\text{H}_2} + x_{\text{O}_2} = 1$$

$$0.4 + x_{\text{O}_2} = 1$$

$$\underline{x_{\text{O}_2} = 0.6}$$

$\textcircled{\text{vii}}$

$$\text{mole} = \frac{\text{given mass}}{\text{Molar mass}}$$

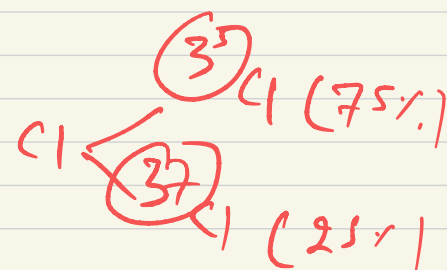
$$2+3 = \frac{100}{M}$$

$$M = \frac{100}{5} = 20g$$

Average atomic mass:-

$$= \frac{a_1 x_1 + a_2 x_2 + \dots}{100}$$

X <



a_1 = molar mass of first atom (isotopes)

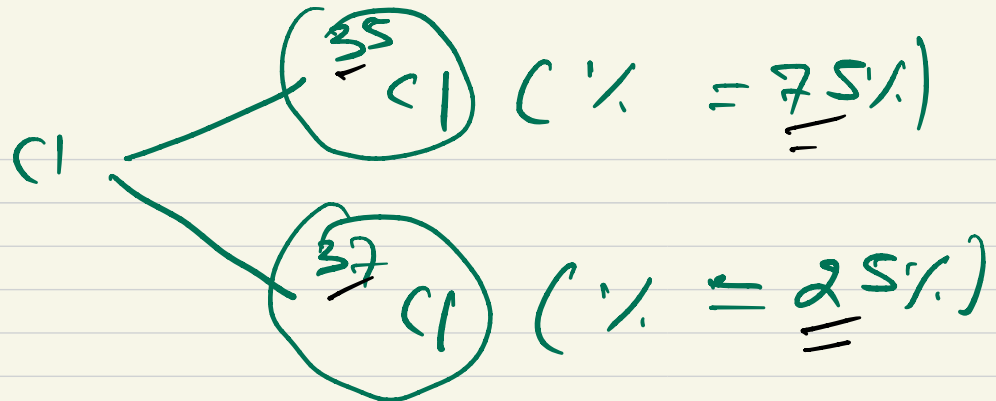
x_1 = mole % of first atom (isotopes)

a_2 = molar mass of second atom (isotopes)

x_2 = mole % of second atom (isotopes)

x_1 = % of Natural abundance

Q.



Find average atomic mass of Cl.

$$\begin{aligned} &= \frac{a_1 \times \%_1 + a_2 \times \%_2}{100} \\ &= \frac{35 \times 75 + 37 \times 25}{100} \\ &= \underline{\underline{35.5 \text{ amu}}} \approx \end{aligned}$$

Q.

Ag atom have two isotopes, $\textcircled{^{109}\text{Ag}}$ and $\textcircled{^{107}\text{Ag}}$.
if the average atomic mass of Ag atom is 108
amu then find % of natural abundance of $\textcircled{^{109}\text{Ag}}$ and $\textcircled{^{107}\text{Ag}}$.

Soln:-

$$\bar{x} = \frac{q_1x_1 + q_2x_2}{100}$$

$$x_1 + x_2 = 100$$

$$= x_1$$

$$= 100 - x_1$$

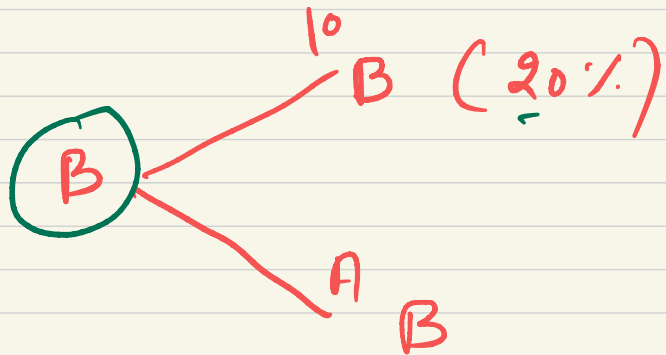
$$108 = \frac{109x_1 + 107(100 - x_1)}{100}$$

$$10800 = 109x_1 + 10700 - 107x_1$$

$$100 = 2x_1$$

$$x_1 = 50\%$$

Q.



Average atomic mass of B = 10.8 amu
then find value of A.

$$\text{AAM} = \frac{q_1 \times 1 + q_2 \times 2}{100}$$

$$10.8 = \frac{10 \times 20 + A \times 80}{100}$$

$$1080 = 200 + 80A$$

$$108 = 20 + 8A$$

$$88 = 8A$$

$$\boxed{A = 11}$$

Average Molar mass :-

$$\text{Average molar mass} = \frac{M_1 X_1 + M_2 X_2 + \dots}{100}$$

M_1 = molar mass of first molecule

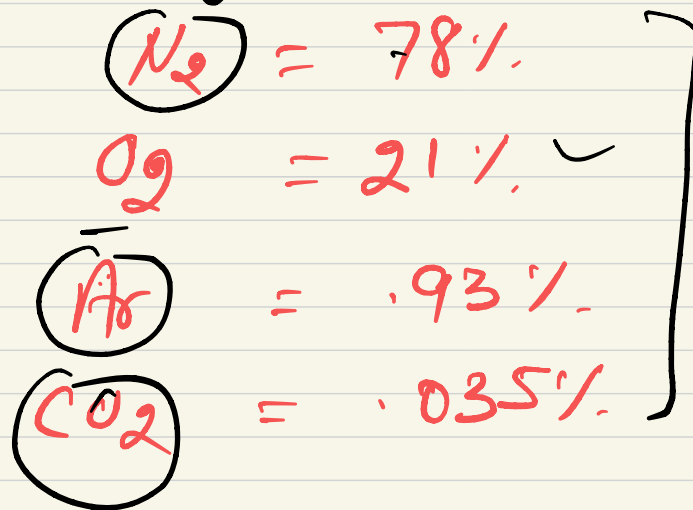
M_2 = " " " second "

X_1 = mole % of first molecule

X_2 = mol%. of second molecule

Q. find average molar mass of air?

air



$\bar{M}_{avg} =$

$$= \frac{28 \times 78 + 32 \times 21 + 40 \times 93 + 44 \times 035}{100}$$

$$= \boxed{28.84\%} \quad \underline{\text{Result}}$$

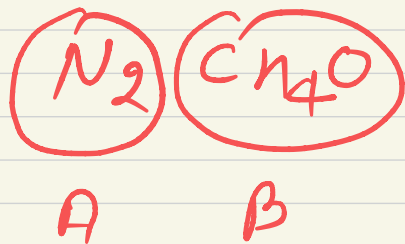
% Composition

$\Rightarrow AB$

Compound written in the form of AB.

$$\% \text{ of A} = \frac{\text{mass of A}}{\text{mass of AB}} \times 100$$

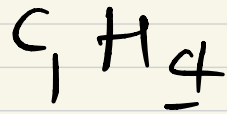
Q. Find % of N in urea. $(\text{NH}_2 - \overset{\text{O}}{\underset{\text{||}}{\text{C}}} - \text{NH}_2)$.



$$M_w = 60 \text{ g}$$

$$\% N = \frac{2 \times 14}{60} \times 100 = \underline{46.6\%}$$

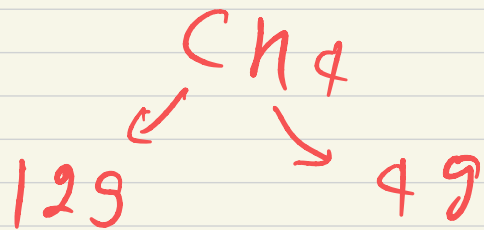
Q. find mole % of C in CH₄ molecule?



$$\% \text{ of C} = \frac{1}{1+4} \times 100 = 20\%$$

$$\% \text{ of H} = 80\%$$

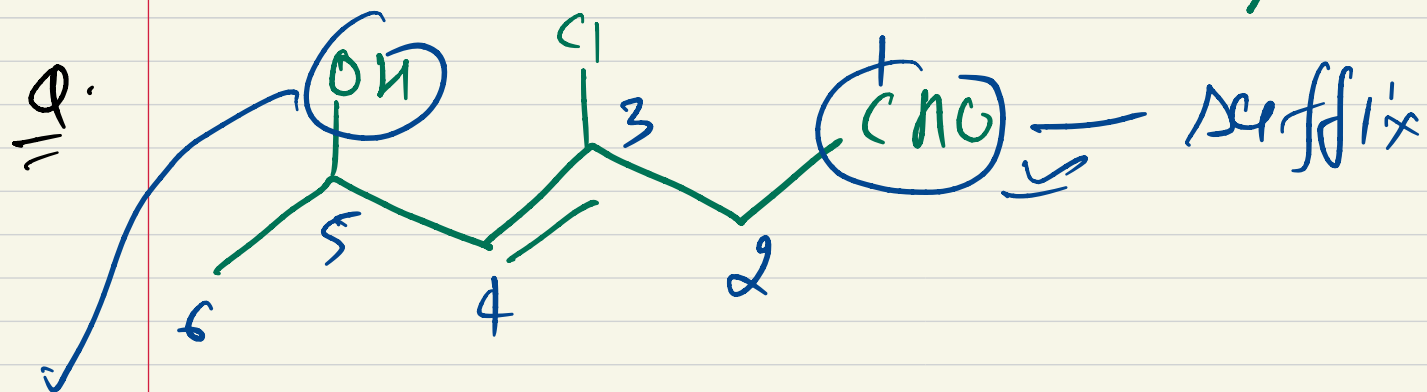
Q. find mass % of carbon in CH₄.



$$\% \text{ of C} = \frac{12}{12+4} \times 100$$

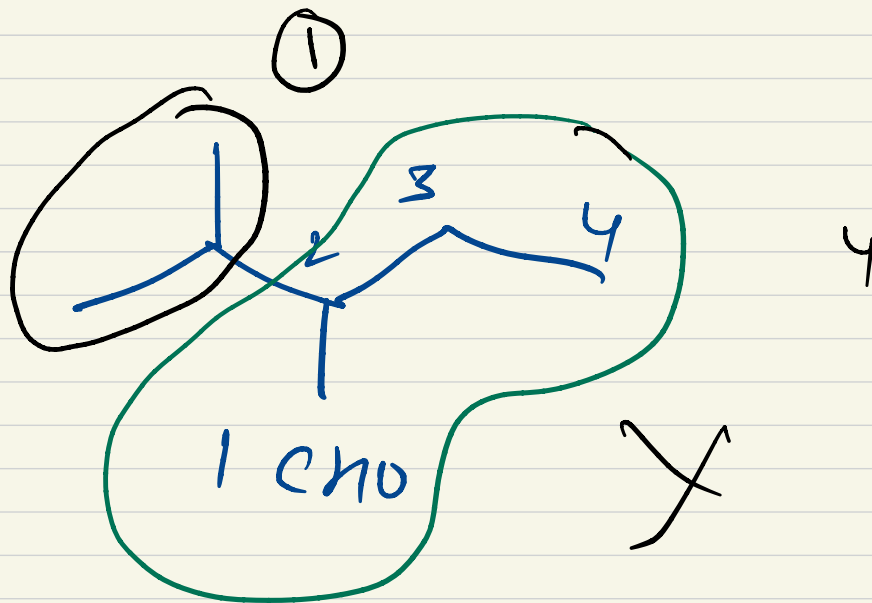
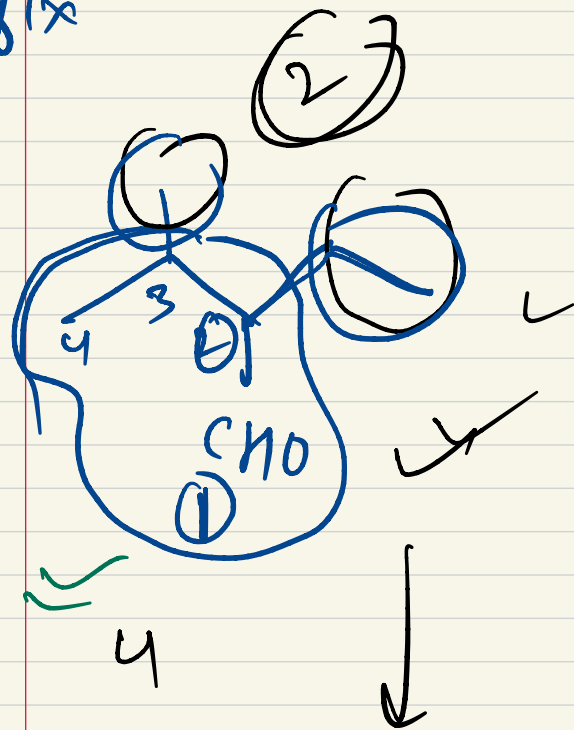
$$= \frac{12}{16} \times 100 = \underline{75\%}$$

3-bromo-5-chloro-2-hydroxypent-1-ol

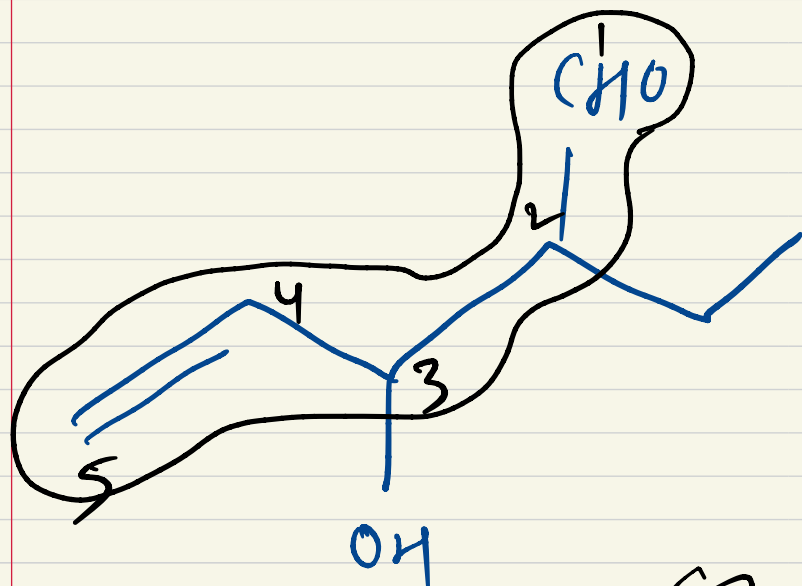


3-chloro-5-hydroxyhex-3-en-1-ol

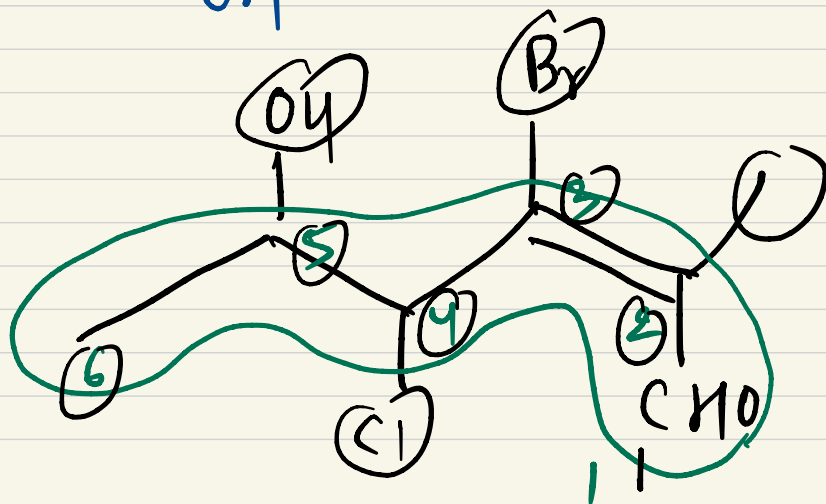
Prefix



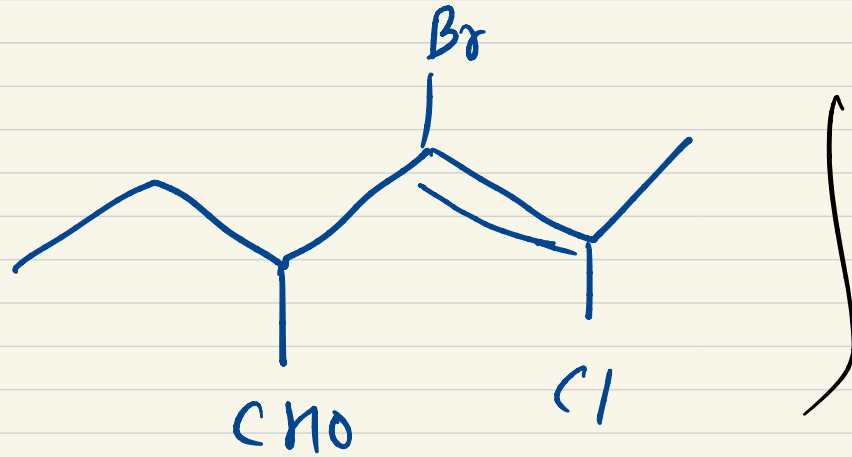
Q. Ethyl-3-methylbut-1-ol —



Q. Ethyl-3-hydroxypent-4-en-1-ol —



3 - Bromo - 4 - chloro - 5 - hydroxy - 2 - methyl hex - 2 - en - 1 - ol



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